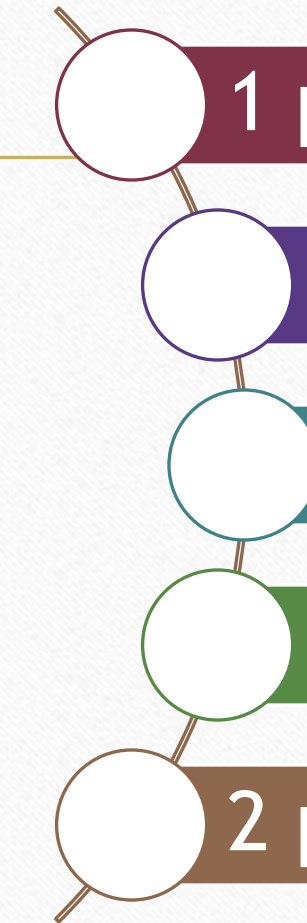


Hypothesis Testing

Topic 5

Hypothesis Testing



1 population mean (large samples)

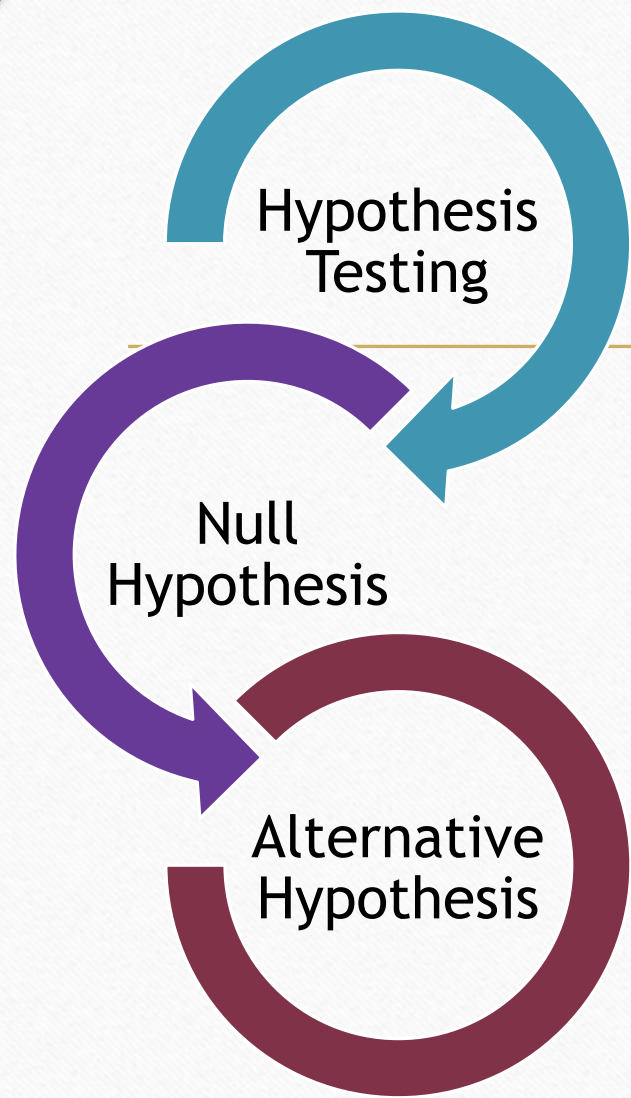
1 population mean (small samples)

2 population mean (large samples)

1 population proportion

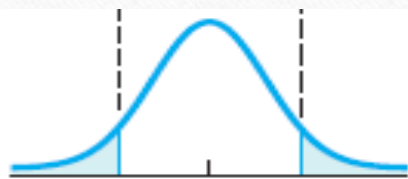
2 population proportion

The Nature of Hypothesis Testing



- Purpose of Hypothesis Testing - to decide whether the null hypothesis should be rejected in favor of the alternative hypothesis.
- Null Hypothesis - The null hypothesis for a hypothesis test concerning a population mean, μ , always specifies a single value for that parameter. Hence, we can express the null hypothesis as $H_0: \mu = \mu_0$ where μ_0 is some number. A hypothesis to be tested. We use the symbol H_0 to represent the null hypothesis.
- Alternative Hypothesis - the choice of the alternative hypothesis depends on and should reflect the purpose of the hypothesis test. Three choices are possible for the alternative hypothesis.

Alternative Hypothesis

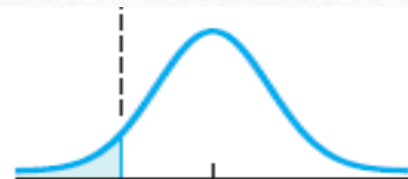


(a) Two tailed

(Two-tailed test)

Deciding the mean is different from a specified value μ_0

$$H_1: \mu \neq \mu_0$$

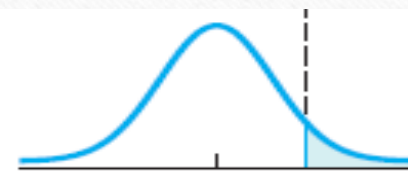


(b) Left tailed

(Left-tailed test)

Deciding the mean is less than a specified value μ_0

$$H_1: \mu < \mu_0$$



(c) Right tailed

(Right-tailed test)

Deciding the mean is greater than a specified value μ_0

$$H_1: \mu > \mu_0$$

Example

A snack company produces a 454g bag of pretzels. As the part of its program, the quality assurance department periodically performs a hypothesis test to decide whether the packaging machine is working properly, that is to decide whether the mean weight of all bags packaged is 454g.

- a) Determine the null hypothesis for the hypothesis test
- b) Determine the alternative hypothesis for the hypothesis test
- c) Classify the hypothesis test as two-tailed, left-tailed or right-tailed.

Example: State the null and alternative hypothesis to be used in this test.

1. The recommended adequate intake of vitamin C for adults is 50 milligrams per day. Suppose that we want to perform a hypothesis test to decide whether the average adult with an income above the poverty level gets more than of 50mg.
2. In a normal situation, a song needs a duration of at most 3.5 minutes to be played. Testing at 5% significance level, do you agree with the claim?
3. A certain calculator has been designed so that the true average lifetime is at least 2 years. Suppose that we want to perform a hypothesis test to verify the claim at 4% significance level.

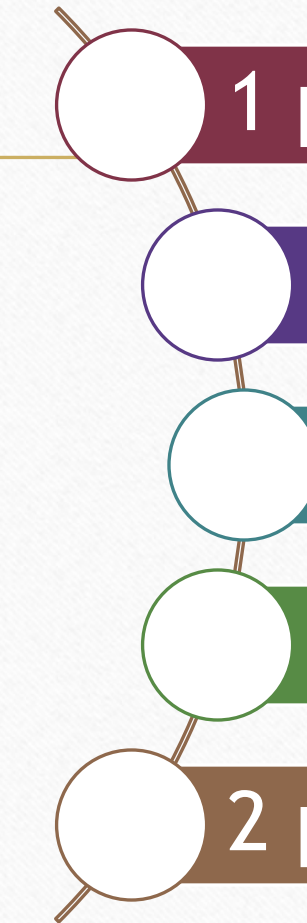
Types of Error

- In the real world we never know if we make an error when conducting a hypothesis test, so we want to keep the probability of making an error small.

Decision	H_0 is True	H_0 is False
Reject H_0		Correct Decision
Accept H_0	Correct Decision	

- Type I Error (False Positive): Rejecting H_0 when in fact H_0 is actually true.
- Type II Error (False Negative): Accepting H_0 when in fact H_0 is actually false.

Hypothesis Testing



1 population mean (large samples)

1 population mean (small samples)

2 population mean (large samples)

1 population proportion

2 population proportion

Hypothesis Testing;

One population mean, large samples

Purpose:

To perform a hypothesis Testing for a population mean, μ

Assumptions;

1. Simple random sample
2. Normal population or large sample
3. σ is known

HT; One population mean, large samples

STEP 1: State the null and alternative hypothesis

The null hypothesis is $H_0: \mu = \mu_0$, and the alternative hypothesis is

STEP 2: Decide on the significant level, α .

STEP 3: Compute the value of the test statistics Z , $z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$

STEP 4: Find the P Value

STEP 5: If $P \text{ Value} \leq \alpha$, reject H_0 ; otherwise, do not reject H_0 .

STEP 6: Interpret the results of the hypothesis test.

Two-tailed test	$H_1: \mu \neq \mu$
Left-tailed test	$H_1: \mu < \mu$
Right-tailed test	$H_1: \mu > \mu$

or

or

Example

In 2000, the mean retail price of all statistic books was RM51.46. This year's retail prices for 40 statistic books are randomly selected, and the sample mean is RM54.89. At the 1% significance level, do the data provide sufficient evidence to conclude that this years' mean retail price of all statistic books has increased? Assume that the standard deviation of prices for this year's statistic books is RM7.61.

Step 1

$$H_0: \mu$$

$$H_1: \mu$$

Step 2

$$\alpha =$$

Step 3

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} =$$

Step 4

Step 5

Step 6

At 1% significance level,
the data provide enough
evidence to conclude that

Exercise

Mean top speed of cheetahs is 60mph. A sample of 35 common cheetahs is selected at random, and the sample mean top speed is 59.526. at the 5% significance level, do the data provide enough evidence to conclude that the mean top speed of all cheetahs differs from 60mph? Assume standard deviation is 3.2mph.

HT; One population mean, small samples

STEP 1: State the null and alternative hypothesis

The null hypothesis is $H_0: \mu = \mu_0$, and the alternative hypothesis is

STEP 2: Decide on the significant level, α .

STEP 3: Compute the value of the test statistics t, $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

STEP 4: Find the critical value(s)

STEP 5: If the value of test statistics falls in the rejection region, reject H_0 ; otherwise do not reject H_0 .

STEP 6: Interpret the results of the hypothesis test.

Two-tailed test	$H_1: \mu \neq \mu_0$
Left-tailed test	$H_1: \mu < \mu_0$
Right-tailed test	$H_1: \mu > \mu_0$

or

or

Critical Value approach for Hypothesis Testing

Rejection region:

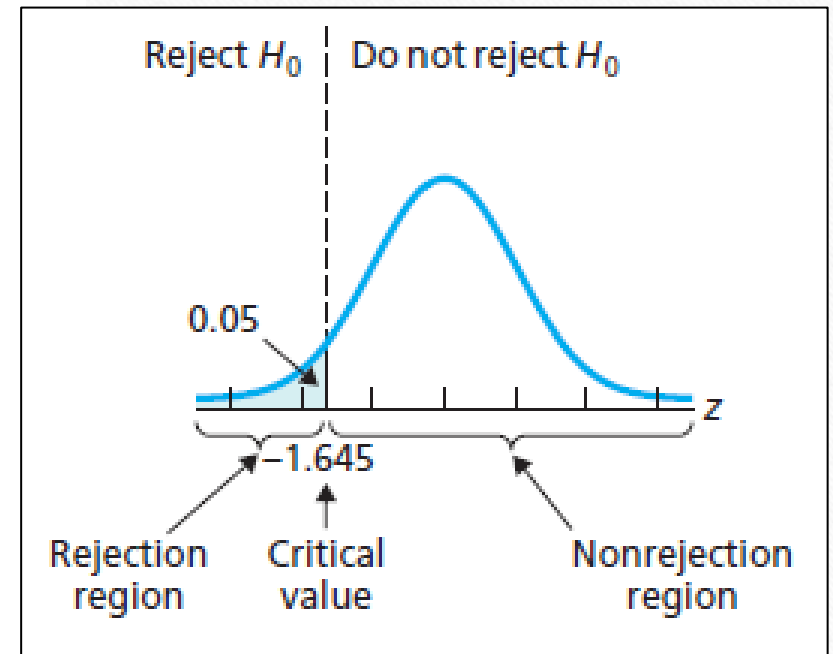
The set of values for the test statistics that leads to rejection of the null hypothesis.

Non rejection region:

The set of values for the test statistics that leads to non-rejection of the null hypothesis.

Critical values:

The values of the test statistics that separate the rejection region and non-rejection regions. A critical value is considered part of rejection region.

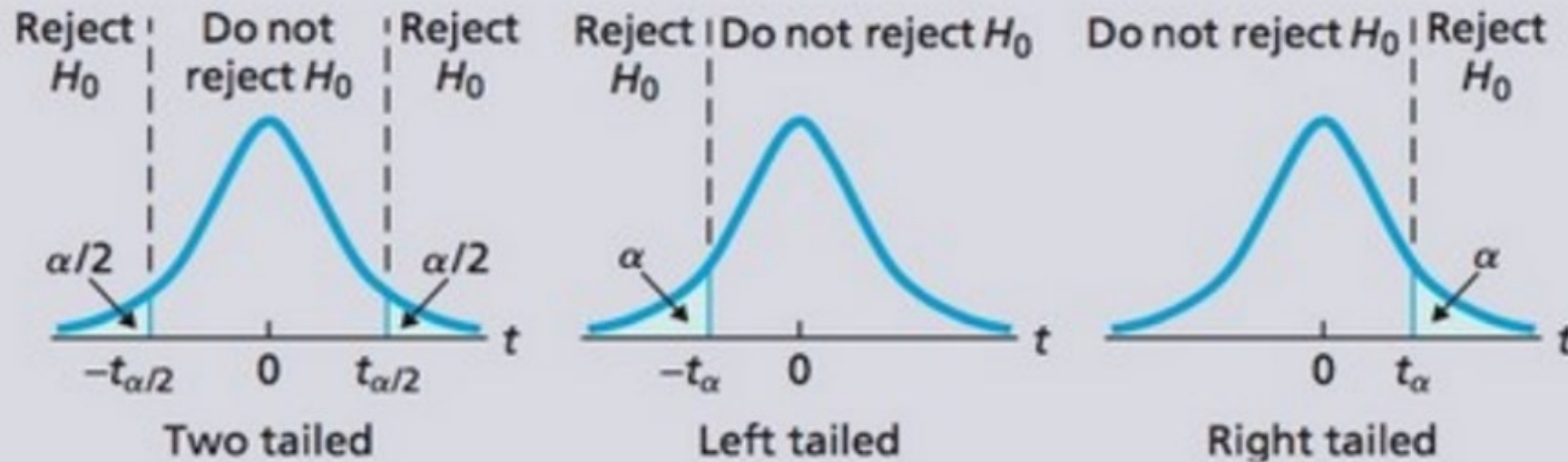


Critical Value approach for Hypothesis Testing

$\pm t_{\alpha/2}$ (Two tailed) or $-t_{\alpha}$ (Left tailed) or t_{α} (Right tailed)

with $df = n - 1$.

Use t-Distribution Table to find the critical value(s).



Example

The recommended adequate intake of calcium for adults is 1000mg per day. A simple random sample of 18 adults gives the average of 947.4mg. At 5% significant level, do the data provide sufficient evidence to conclude that the mean calcium intake of all adults is less than 1000mg? Assume $\sigma = 188$ mg.

Step 1

$$H_0: \mu$$

$$H_1: \mu$$

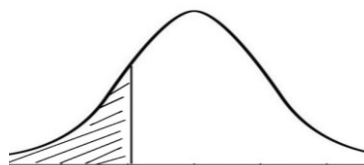
Step 2

$$\alpha = 5\%$$

Step 3

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} =$$

Step 4



Step 5

Step 6

At 5% significance level,

Exercise

A psychologist claims that the mean age at which children start walking is 12.5 months. Carol wants to check if this claim is true. She took a random sample of 18 children and found that the mean age at which these children started walking was 12.9 months with standard deviation of 0.8 month. Using 1% of significance level, can you conclude that the mean age at which children start walking is different from 12.5 months? Assume that the ages at which all children start walking have an approximately normal distribution.

Hypothesis Testing; One population proportion

Definition

Consider a population in which each member either has or does not have a specified attribute.

Terminology

1. Population Proportion, p :

The proportion (percentage) of the entire population that has the specified attribute

2. Sample proportion, $\hat{p}$:

The proportion (percentage) of a sample from the population that has the specified attribute.

Hypothesis Testing; One population proportion

Purpose:

To perform a hypothesis Testing for a population proportion, p

A sample proportion, $\hat{p}$ is computed by using the formula

$$\hat{p} = \frac{x}{n}$$

Where x denotes the number of members in the sample that have the specified attribute and n denotes the sample size

Assumptions:

1. Simple random sample
2. Both np_0 and $n(1 - p_0)$ are 5 or greater

HT; One population proportion

STEP 1: State the null and alternative hypothesis

The null hypothesis is $H_0: p = p_0$, and the alternative hypothesis is

STEP 2: Decide on the significant level, α .

STEP 3: Compute the value of the test statistics Z,

STEP 4: Find the P-Value

STEP 5: If ***P Value*** $\leq \alpha$, reject H_0 ; otherwise, do not reject H_0 .

STEP 6: Interpret the results of the hypothesis test.

Two-tailed test	$H_1: p \neq p_0$
Left-tailed test	$H_1: p < p_0$
Right-tailed test	$H_1: p > p_0$

or

or

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

Where $\hat{p} = \frac{x}{n}$

Example

In late January 2019, Gallup Company conducted a national poll of 1053 U.S. adults that asked their views on an economic stimulus plan whether to favor or oppose the passage. Of those sampled, 548 favored passage. At the 5% significance level, do the data provide sufficient evidence to conclude that a majority of U.S. adults favoured passage?

Exercise

A commonly prescribed drug for relieving nervous tension is believed to be only 60% effective. Experimental results with a new drug administered to a random sample of 100 adults who were suffering from nervous tension show that 70 received relief. Is this sufficient evidence to conclude that the new drug is superior to the one commonly prescribed? Use a 0.05 level of significance.

HT; Two population proportion

STEP 1: State the null and alternative hypothesis

The null hypothesis is $H_0: p_1 = p_2$, and the alternative hypothesis is

STEP 2: Decide on the significant level, α .

STEP 3: Compute the value of the test statistics Z,

STEP 4: Find the Pvalue(s)

STEP 5: If ***P Value*** $\leq \alpha$, reject H_0 ; otherwise, do not reject H_0 .

STEP 6: Interpret the results of the hypothesis test.

Two-tailed test	$H_1: p_1 \neq p_2$
Left-tailed test	$H_1: p_1 < p_2$
Right-tailed test	$H_1: p_1 > p_2$

or

or

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}_p(1 - \hat{p}_p)} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where

$$\hat{p}_p = \frac{x_1 + x_2}{n_1 + n_2}$$

Example

In a study of time of arrival for work, it was found that there were 25 people out of 300 in a sales department who were lacking in punctuality. The figures for the accounting department indicated that 43 out of a staff of 600 were not punctual. Do the proportion of employees who are lacking in punctuality in the sales department higher than accounting department at 1% level of significance?

Exercise

In a sample of 600 men from a certain city, 450 men are found to be smokers. In a sample of 900 men from another city 630 are found to be smokers. Do the data indicate that two cities are significantly different with respect to prevalence of smoking habit among men at 10% level of significance?

HT; Two population mean

STEP 1: State the null and alternative hypothesis

The null hypothesis is $H_0: \mu_1 = \mu_2$, and the alternative hypothesis is

STEP 2: Decide on the significant level, α .

STEP 3: Compute the value of the test statistics Z,

STEP 4: Find the P value(s)

STEP 5: If $P \text{ Value} \leq \alpha$, reject H_0 ; otherwise, do not reject H_0 .

STEP 6: Interpret the results of the hypothesis test.

Two-tailed test	$H_1: \mu_1 \neq \mu_2$
Left-tailed test	$H_1: \mu_1 < \mu_2$
Right-tailed test	$H_1: \mu_1 > \mu_2$

or

or

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Example

A researcher conducts salary studies of lecturers in private university and public university. A sample of 50 lecturers were interviewed, and the mean salary is RM3650. A sample of 60 lecturers from public university were interviewed and their mean salary is RM3890. It is known that the standard deviation for private university and public university are RM500.5 and RM320 respectively. Decide whether the mean salaries of lecturers in private university is less than the lecturers in public university at 5% level of significance

Exercise

The number of components produced by a particular employee was recorded each day for a random sample of 30 recent days. The sample mean of the number produced was 48.2. The number of components produced by the same employee was recorded each day for a random sample of 60 days last year and the sample mean was 45.9. Given that the population standard deviation for current year and last year were 3.5 and 4.5 respectively. Test whether there is evidence of a significant increase in the mean daily production by employee at 1% level of significance.

End of topic 5

Population	Mean	Proportion
ONE	$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \quad \text{or} \quad t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$	$Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$
TWO	$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}_p(1 - \hat{p}_p)} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ where $\hat{p}_p = \frac{x_1 + x_2}{n_1 + n_2}$